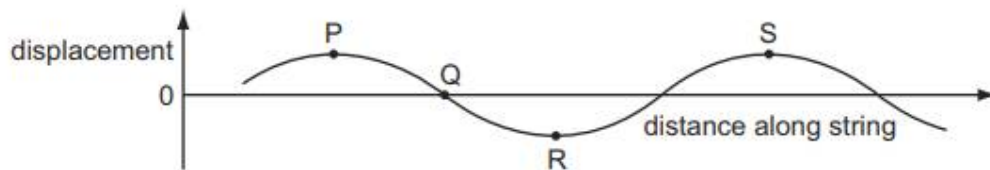


- 17 The string shows the shape at a particular instant of part of a progressive wave travelling along a string.



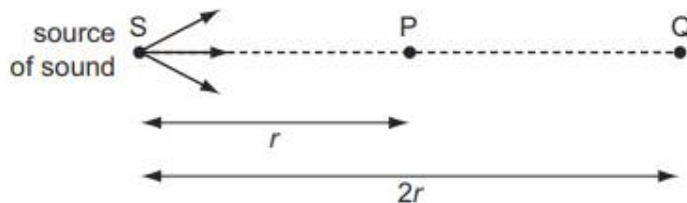
Which statement about the motion of the points along the string is correct?

- A The speed at point P is maximum.
- B The displacement at point Q is always zero.
- C The energy at point R is entirely kinetic.
- D The acceleration at point S is maximum.

Ans: D

At Point S, the displacement of the particle is maximum. Hence, by using $a = -\omega^2 x$, the acceleration will also be at a maximum.

- 18 A point source of sound is placed at point S.



Ans: D

Amplitude is inversely proportional to the distance from the source.

The air molecules at P, a distance r from S, oscillate with an amplitude of $8.0 \mu\text{m}$. Point Q is situated at a distance $2r$ from S.

What is the amplitude of oscillation of air molecules at point Q?

- A** $1.4 \mu\text{m}$ **B** $2.0 \mu\text{m}$ **C** $2.8 \mu\text{m}$ **D** $4.0 \mu\text{m}$

- 19 Two loudspeakers are emitting sound of wavelength λ in all directions. They are in phase with